The Department Name is the Department of Software Engineering under the Faculty of Information & Communication Technology (FICT), and the program is accredited by the Pakistan Engineering Council (PEC). The core focus of the program combines theoretical and practical software engineering principles, encompassing a systematic, disciplined, and quantifiable approach to the design, development, operation, and maintenance of large-scale, reliable, efficient, and economical software systems. Historically, the department has been offering a four-year undergraduate software engineering degree since 2012 to meet growing national and international market demands. Specialization fields include requirements engineering, quality engineering, software testing, and project management. The vision is to attain excellence in education and research at the national and international level and produce technically, professionally, and ethically sound graduates. The mission is to produce next-generation Software Engineering graduates who are fully equipped with concrete software engineering knowledge, real-world problem-solving skills, research abilities, excellent leadership qualities, and entrepreneurial abilities required for their future careers and serving their communities. Admission requirements for the Bachelor of Science in Software Engineering (BSSE) include an academic background of F.Sc (Pre-Engineering) or ICS with Mathematics and Physics from any recognized board, or equivalent qualification, with at least 60% marks, or alternatively, a Diploma of Associate Engineer (DAE) in a relevant field with a minimum of 60% marks. Degree requirements include a total of 132 credit hours across 42 courses plus the Final Year Design Project (FYDP Part I & II) with a minimum academic standing of a CGPA greater than or equal to 2.0. The department targets training students across all aspects of the software development lifecycle to achieve Program Educational Objectives (PEOs). These PEOs include utilizing critical-thinking and engineering abilities to excel in the domain of Software Engineering, acting as professional, ethical, and knowledgeable citizens, and demonstrating leadership while working collaboratively in diverse teams and organizations.

The curriculum plan follows a semester schema where credit hour listings follow the format of Theory-Lab. In the 1st Semester, courses include IT-102 Information and Communication Technology (ICT) with prerequisite NIL and 2-1 credit hours, SE-2XX Software Engineering Fundamentals with prerequisite NIL and 3-0 credit hours, ENG-161 Reading and Writing Skills with prerequisite None and 3-0 credit hours, PHY-205 Applied Physics with prerequisite NIL and 2-1 credit hours, HUM-138 or HUM-1XX Psychology or Islamic Studies and Ethics with prerequisite NIL/NIL and 2-0 or 2-0 credit hours, and XX-1XX Occupational Health and Safety with prerequisite NIL and 1-0 credit hours. In the 2nd Semester, courses include CS-114 Programming Fundamentals with prerequisite NIL and 3-1 credit hours, ENG-361 Communication & Presentation Skills with prerequisite NIL and 3-0 credit hours, MATHA-234 Discrete Structures with prerequisite NIL and 3-0 credit hours, MATHP-118 Calculus & Analytical Geometry with prerequisite NIL and 3-0 credit hours, and CS-119 or HUM-102 Web Development (WD) or Pakistan Studies & Global Perspective with prerequisite NIL/NIL and 2-1 or 2-0 credit hours.

In the 3rd Semester, courses include CS-212 Object-Oriented Programming with a prerequisite of Programming Fundamentals and 3-1 credit hours, STAT-101 Probability & Statistics with prerequisite NIL and 3-0 credit hours, CE-2XX Computer Architecture and Logic Design with prerequisite NIL and 3-1 credit hours, MATHP-111 Linear Algebra with prerequisite NIL and 3-0 credit hours, and CE-302 SE Elective I (Web Engineering (WE)) with a prerequisite of Web

Development and 2-1 credit hours. In the 4th Semester, courses include CS-214 Data Structures and Algorithms with a prerequisite of Object-Oriented Programming and 3-1 credit hours, SE-231 Software Design & Architecture with a prerequisite of Software Engineering Fundamentals and 2-1 credit hours, IT-3XX Human-Computer Interaction with prerequisite NIL and 2-1 credit hours, XX-XXX SE Elective - II with prerequisite NIL and 3-0 credit hours, and MGMT-206 or MATHA-213 Principles of Management or Complex Variables & Transform with prerequisite NIL/NIL and 3-0 or 2-0 credit hours. In the 5th Semester, courses include SE-332 Software Construction and Development with a prerequisite of Software Design & Architecture and 2-1 credit hours, MGMT-2XX Entrepreneurship and Marketing with prerequisite NIL and 2-0 credit hours, XX-XXX SE Elective - III with prerequisite NIL and 3-0 credit hours, and CS-332 or CS-341 Database Systems or Operating Systems with prerequisite NIL/NIL and 3-1 or 3-1 credit hours. In the 6th Semester, courses include SE-321 Software Quality Engineering with a prerequisite of Software Engineering Fundamentals and 3-0 credit hours, TE-307 Computer Networks with prerequisite NIL and 3-1 credit hours, MATHA-216 Numerical Analysis with prerequisite NIL and 2-1 credit hours, XX-XXX MDEE-I with prerequisite NIL and 3-0 credit hours, and HUM-266 Technical & Business Writing with prerequisite NIL and 3-0 credit hours. In the 7th Semester, courses include SE-352 Software Project Management with prerequisite NIL and 3-0 credit hours, IT-321 Information Security with prerequisite NIL and 3-0 credit hours, SE-491L Senior Design Project - I (FYDP-I) with prerequisite NIL and 0-3 credit hours, XX-XXX SE Elective - IV with prerequisite NIL and 3-0 credit hours, and CS-4XX Design and Analysis of Algorithms with a prerequisite of Data Structures and Algorithms and 3-0 credit hours. Finally, in the 8th Semester, courses include XX-XXX MDEE-II with prerequisite NIL and 3-0 credit hours, SE-492L Senior Design Project - II (FYDP-II) with a prerequisite of Senior Design Project - I and 0-3 credit hours, XX-XXX or XX-XXX SE Elective - V or SE Elective VI with prerequisite NIL/NIL and 3-0 or 3-0 credit hours, and SE-471 Formal Methods in Software Engineering with prerequisite NIL and 3-0 credit hours.